



FutureTech HackFest 2026

# FarmGuard AI

Smart Crop Disease Detection Platform

Theme: **Innovating Tomorrow with Emerging Technologies**

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## The Problem

**20–40%**

Annual crop loss in India

**70%**

Farmers are small/marginal

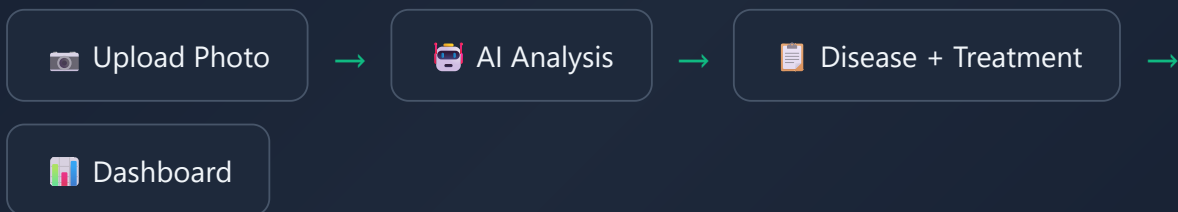
**Days**

To get expert diagnosis

- ▶ Plant diseases destroy crops before farmers can identify them
- ▶ Rural areas lack access to agricultural experts
- ▶ Traditional diagnosis is slow, costly, and reactive
- ▶ Late treatment leads to financial loss and food waste

## Our Solution — FarmGuard AI

A **full-stack AI web platform** that lets farmers photograph a crop leaf and get instant disease diagnosis with treatment advice.



No expensive hardware — just a phone and internet connection.

## Key Features



### Secure Login

Farmer authentication with protected dashboard access



### AI Disease Scanner

Upload leaf image → instant disease identification



### Confidence Score

AI confidence % + severity (Low / Medium / High)



### Treatment Guide

Actionable treatment & prevention recommendations



### Analytics Dashboard

Scan history, risk stats, and crop health tracking



### Mobile Friendly

Works on any phone, tablet, or computer

## How It Works

### User Flow

- ▶ Farmer logs into FarmGuard AI
- ▶ Selects crop type (Tomato, Rice, etc.)
- ▶ Uploads photo of affected leaf
- ▶ AI model analyzes the image
- ▶ Results: disease, confidence, severity
- ▶ Treatment recommendation displayed
- ▶ History saved on dashboard

### System Architecture

**Frontend** — Next.js + React + Tailwind

**Backend** — Node.js API Routes

**AI/ML** — Hugging Face Vision Transformer

**Deploy** — Vercel Cloud Platform

**Auth** — JWT Session Authentication

# Technology Stack

**Next.js 14**  
Framework

**React 18**  
UI Library

**TypeScript**  
Language

**Tailwind CSS**  
Styling

**Node.js**  
Runtime

**Hugging Face**  
ML Model

**Vercel**  
Deployment

**Git / GitHub**  
Version Control

**Cursor AI**  
Development

Languages: **TypeScript** · **JavaScript** · **HTML** · **CSS**

## Innovation & Uniqueness

- ▶ **Accessibility First** — Simple UI for rural farmers, no tech skills needed
- ▶ **Instant Results** — Photo to diagnosis in seconds vs. days with experts
- ▶ **Zero Hardware Cost** — Phone camera only, no IoT sensors required
- ▶ **Full-Stack Platform** — AI + Auth + Dashboard in one integrated app
- ▶ **Treatment Guidance** — Not just detection, but actionable advice
- ▶ **Sustainable Impact** — Reduces crop waste & pesticide overuse
- ▶ **Cloud-Ready** — Scalable architecture for millions of farmers

## Live Demo

### Links

GitHub:

[github.com/gowthamcodes225/farmguard-ai](https://github.com/gowthamcodes225/farmguard-ai)

Live Demo:

[farmguard-ai.vercel.app](https://farmguard-ai.vercel.app)

### Demo Login

Email: [farmer@farmguard.ai](mailto:farmer@farmguard.ai)

Password: [demo123](#)

## Demo Steps

1. Login



2. Upload Leaf Photo



3. Analyze Disease



4. View Dashboard



## Future Scope

- ▶ Android / iOS mobile application
- ▶ IoT soil sensor integration (moisture, pH)
- ▶ Multi-language support (Tamil, Hindi)
- ▶ Government & NGO farmer outreach
- ▶ Regional disease outbreak heatmaps
- ▶ Market price & sell-timing recommendations
- ▶ Offline mode for low-connectivity areas
- ▶ Integration with agri-input suppliers



# Thank You!

FarmGuard AI — Protecting Crops with AI

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GitHub: [github.com/gowthamcodes225/farmguard-ai](https://github.com/gowthamcodes225/farmguard-ai)

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